



Clyde Rempillo, M.Sc., B.C.S.
Senior Software Engineer | AI/ML Architect | Enterprise Solutions Lead
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Professional Summary

Senior software engineer and AI/ML architect with 9+ years of professional industry experience and 12+ years of hands-on software development. Progressed from independent development and a software engineering internship into full-time engineering, enterprise AI/ML, architecture, and technical leadership roles. Corporate and solution-delivery experience includes Splice, Google, ADP, Hiverlab, Toronto Metropolitan University, and UAE technology organizations. Proven ownership across requirements discovery, system architecture, full-stack engineering, commercial software, agentic systems, ERP/CRM implementation, cloud deployment, DevOps, QA, and production support. Experienced in translating operational challenges into secure, scalable software across logistics, smart cities, HR/payroll, accounting, education, healthcare, and industrial domains. Hands-on with Odoo, Zoho, Microsoft Azure, AWS, and GCP, with a record of leading distributed teams and communicating with executive, product, operations, and engineering stakeholders.

Core Competencies

- Software Engineering:** System design, software architecture, object-oriented programming, design patterns, SOLID, REST APIs, microservices, event-driven workflows, data modeling, SDLC, Agile/Scrum, TDD, code review, QA, and production support.
- Languages & Frameworks:** Python, Java, JavaScript, TypeScript, C/C++, C#, PHP, SQL, HTML/CSS, React, Next.js, Node.js, Express/Fastify, Flask, FastAPI, Bootstrap, JUnit, and WordPress.
- AI/ML & Agentic Systems:** LLM applications, AI agents, RAG, prompt engineering, tool/function calling, TensorFlow, PyTorch, predictive analytics, feature engineering, model evaluation, MLOps, computer vision, IoT, CPS, digital twins, and simulation.
- ERP/CRM & Enterprise Integration:** Odoo ERP, Odoo CRM, Accounting, Invoicing, Sales, HR, Employees, Recruitment, Zoho CRM, Zoho One, REST APIs, webhooks, field mapping, data synchronization, migration planning, RBAC, workflow automation, and operational reporting.
- Cloud, Data & DevOps:** Microsoft Azure, AWS, GCP/Firebase, PostgreSQL, MySQL, Firestore, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git, Linux/Unix, CI/CD, monitoring, and cloud-native deployment.
- Leadership & Delivery:** Technical strategy, architecture governance, engineering leadership, mentoring, roadmap planning, estimation, backlog and sprint management, stakeholder workshops, vendor/client coordination, documentation, and executive communication.

Professional Experience

- Head of AI Software Solutions | Enterprise AI & Automation

Aug 2025 - Present

Yo Technology Co. Dubai, UAE

- Set the technical direction for internal platforms, commercial software products, and client-facing AI solutions, leading delivery from discovery and architecture through implementation, QA, deployment, and operational handover.
 - Implemented and extended **Odoo as the company’s internal ERP**, supporting CRM, sales, HR, recruitment, employee administration, accounting, invoicing, approvals, reporting, and cross-functional operational workflows.
 - Architected modular agentic systems and automated workflows for HR and accounting, connecting LLMs, business rules, Odoo/Zoho data, APIs, document processing, notifications, and human approval checkpoints.
 - Designed reusable ERP/CRM integration frameworks for Odoo and Zoho, enabling data synchronization, lead management, employee workflows, invoice processing, operational reporting, exception handling, and role-based access.
 - Led development of commercial conversational-AI capabilities, including a real-time voice sales agent built with LiveKit, React, OpenAI, AssemblyAI, and ElevenLabs.
 - Partnered with executive, product, finance, HR, operations, and engineering stakeholders to prioritize roadmaps, manage delivery risk, and translate business requirements into scalable production software.
- Senior Software & AI/ML Architect | Enterprise Digital Twins

Jan 2024 - Aug 2025

Hiverlab Co. / Nasser Al Jaghoub Trading LLC Dubai, UAE

- Led solution architecture and technical delivery for client-facing AI, IoT, CPS, simulation, and digital-twin programs across smart-city, logistics, built-environment, and industrial domains.
 - Architected the **NEOM Smart Lift System**, connecting IoT and equipment data with digital-twin models and predictive workflows for monitoring, operational visibility, and maintenance planning.
 - Developed architecture and automation pipelines for the **DB Schenker Warehouse Digital Twin**, supporting process simulation, data integration, scenario analysis, and logistics optimization.
 - Directed implementation across Middle East and Singapore-based teams, converting stakeholder requirements into system designs, delivery roadmaps, interfaces, acceptance criteria, and engineering work packages.
 - Provided architecture governance and hands-on technical guidance on scalability, security, maintainability, integration design, testing, and production readiness.
- Software Engineer | Internal Platforms & Automation

Dec 2022 - Jan 2024

Toronto Metropolitan University - Lincoln Alexander School of Law Toronto, ON, Canada

- Designed and delivered internal applications and staff systems supporting HR, faculty administration, operational workflows, reporting, and knowledge access.
- Built responsive web applications and dashboards using Node.js, Bootstrap, JavaScript, and Firestore/Firebase, improving the consistency and accessibility of administrative information.
- Owned requirements gathering, implementation, testing, documentation, deployment support, and iterative improvements in collaboration with IT, faculty, and administrative stakeholders.
- Applied role-based access, secure data-handling practices, maintainable code structures, and operational support processes to deployed solutions.

Full-Stack Software Engineer | Enterprise HR, Payroll & CRM

Oct 2021 - Dec 2022

ADP Technology Inc.

Etobicoke, ON, Canada

- Built and maintained enterprise web applications and internal tools supporting HR, payroll, CRM, customer operations, and data-driven business workflows in a large corporate environment.
- Developed responsive front-end features and backend services using React, Node.js, JavaScript, HTML/CSS, REST APIs, Firestore, and SQL-based data integrations.
- Integrated application workflows with ERP/CRM data sources and Microsoft Azure-hosted services, supporting secure data exchange, workflow automation, deployment, and operational troubleshooting.
- Worked within Agile delivery practices across requirements refinement, estimation, sprint execution, code review, testing, defect resolution, release support, and technical documentation.
- Collaborated with product, operations, QA, and engineering stakeholders to convert enterprise requirements into reliable, maintainable, and user-focused software.

Junior AI/ML Engineer | Machine Learning

Sep 2020 - Oct 2021

Google Inc. - Machine Learning Department

Toronto, ON, Canada

- Developed machine-learning workflows in a cross-functional corporate environment using Python, TensorFlow, GCP, Firebase, and cloud-based data services.
- Built reproducible pipelines for data preprocessing, feature engineering, model training, evaluation, validation, and performance analysis for predictive use cases.
- Integrated model outputs into cloud-connected applications and automated workflows, working with software and product stakeholders to operationalize ML capabilities.
- Produced technical documentation, communicated experimental results, and supported iterative improvements to model quality, reliability, and maintainability.

Software Developer Intern → Junior Software Developer

2017 - Aug 2020

Splice - Outsourced Software Development

Windsor, ON, Canada

- Joined through a software development internship and progressed into a full-time junior developer role, gaining first professional industry experience delivering outsourced software solutions for client organizations.
- Contributed to full-stack web applications, internal business tools, integrations, and database-backed features using JavaScript, React, Node.js, PHP, SQL, HTML/CSS, and REST APIs.
- Worked across requirements analysis, feature implementation, debugging, testing, Git-based version control, documentation, release support, and maintenance across multiple client engagements.
- Collaborated with senior engineers, project leads, and client stakeholders to translate business needs into maintainable software while developing professional engineering and delivery practices.

Independent Software Developer & IT Consultant

2014 - 2017

Independent / Client Projects

Canada / Remote

- Started with websites, IT support, and workflow automation, then progressed into full-stack applications, database-backed systems, cloud deployments, and early AI/ML prototypes for small-business and independent clients.
- Delivered and supported end-to-end solutions using JavaScript, React, Node.js, Python, PHP, SQL, Firebase, WordPress, AWS, and Azure, covering requirements, implementation, testing, hosting, documentation, and post-deployment support.

Selected Enterprise & AI Projects

Odoo ERP, Agentic HR & Accounting Automation | Odoo, Zoho, Python, REST APIs

2025 - Present

- Implemented Odoo as an internal ERP and built agent-assisted workflows spanning CRM, recruitment, employee administration, accounting, invoicing, approvals, document processing, reporting, data synchronization, access control, and exception handling.

STL4IoT Statechart Template Library | IoT, CPS, Model-Driven Engineering

2020 - 2022

- Developed reusable statechart templates for IoT and cyber-physical systems; work contributed to a peer-reviewed ANNSIM 2023 publication.

Education & Credentials

Master of Science in Software Engineering

2022

Toronto Metropolitan University

Toronto, ON, Canada

- GPA: 3.92/4.00; Ryerson Graduate Fellowship. Thesis focused on model-driven engineering for IoT, cyber-physical systems, and digital twins.

Bachelor of Computer Science; Minor in Mathematics & Statistics

2020

University of Windsor

Windsor, ON, Canada

- GPA: 3.2/4.0; University of Windsor Scholarship recipient.

Certifications: Google and IBM AI/ML credentials; CompTIA Security+; CompTIA Network+